

Your grade: 100%

Next item →

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1.

What are the benefit(s) to using modules on a system versus installing software in your workspace directory (i.e., /projects or /work)?

1 / 1 point

If you have conflicting compiler versions, a module system will allow you (or the system administrator) to manage that effectively

Modules load automatically when a new version comes out

Software you install from your workspace directory cannot be accessed by any of your collaborators

All of the above

Nice work

One of the greatest benefits of using a modules system is that it manages any dependencies or versions of the software

2.

In the videos, we discussed why HPC centers might use allocations to manage usage on a system. But why might allocations processes be good for the user?

1 / 1 point

It forces the user to plan out their compute usage for their project

It might promote more efficient use of code, resulting in quicker execution times

Going through the allocations process can help the user better understand how the system works

All of the above

Nice work

3.

Which job would be best run on the compute nodes?

1 / 1 point

Short jobs that do not take up much CPU

“Heavy” jobs that require a lot of CPU

Jobs that require a small amount of memory

All of the above

Nice work

4.

Which job would be best run on a login node?

1 / 1 point

Short jobs that do not take up much CPU

“Heavy” jobs that require a lot of CPU

Jobs that require a small amount of memory

None of the above

Nice work

No jobs should be run on login nodes

See the videos on software and job submissions with Slurm to review

5.

Where should one type the command to load software needed to execute code? For example, if the batch script I'm running on the HPC system needs Python, where should I type the commands to load the Python module?

1 / 1 point

From the command line on the login node

In the batch script that you are executing

From the command line on the compute node

From the command line on the compile node

Nice work

The batch script will be running on the compute node, and if you put it in the batch script then you will have it for future reuse.

See the video on HPC architecture to review

6.

After this class, you get a job at an HPC center, and your primary role is to design the next supercomputer. This involves purchasing the hardware and software needed, as well as figuring out the infrastructure in which to hold it. What, of the items listed below, are your biggest limiting factors in how fast parallel code will run on the system? The main workload this HPC center does is single node computing.

1 / 1 point

The number of nodes and cores

The speed of the network and the number of cores

The number of nodes and the speed of the network

The amount of storage and the number of cores.

Nice work

The speed of the interconnect is important for fast calculations, and the number of cores are important for power.

See the videos on job submission with Slurm review

7.

Let's say that I ask for 10 hours of wall time on my job, but it actually takes 10 hours and 30 minutes to complete. What will happen?

1 / 1 point

The job will run for 10 hours and stop

The job will run for 10 hours and 30 minutes and will stop

Slurm will recognize that your job will go over time and will ask if you want to extend the time

Slurm won't start the job won't start because it will know your job will take more time

Nice work

Since you've only asked for ten hours of time, Slurm will only allocate ten hours of resources for you

https://www.coursera.org/learn/introduction-high-performance-computing/assignment-submission/CYwNt/week-2-quiz/view-feedback

1/1